

Airline Safety Improvement Through Experience with Near-Misses: A Cautionary Tale

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In recent years, the U.S. commercial airline industry has achieved unprecedented levels of safety, with the statistical risk associated with U.S. commercial aviation falling to 0.003 fatalities per 100 million passengers. But decades of research on organizational learning show that success often breeds complacency and failure inspires improvement. With accidents as rare events, can the airline industry continue safety advancements? This question is complicated by the complex system in which the industry operates where chance combinations of multiple factors contribute to what are largely probabilistic (rather than deterministic) outcomes. Thus, some apparent successes are realized because of good fortune rather than good processes, and this research intends to bring attention to these events, the **near-misses**. The processes that create these near-misses could pose a **threat** if **multiple contributing** factors combine in adverse ways without the intervention of good fortune. Yet, near-misses (if recognized as such) can, theoretically, offer a **mechanism for continuing safety improvements**, above and beyond learning gleaned from observable failure. **We test whether or not this learning is apparent in the airline industry**. Using data from 1990 to 2007, fixed effects Poisson regressions show that airlines **learn from accidents** (their own and others), and from one **category of near-misses**—those where the **possible dangers are salient**. **Unfortunately**, airlines do not improve following near-miss incidents when the **focal event has no clear warnings of significant danger**. Therefore, while airlines need to and can learn from *certain* near-misses, we conclude with recommendations for improving airline learning from *all* near-misses.

KEY WORDS: Commercial aviation; near-misses

1. INTRODUCTION

Traveling by large U.S. commercial airline is one of the safest modes of transportation in the world with an estimated 0.003 fatalities per 100 million passenger miles.⁽¹⁾ Between 1998 and 2008, the fatality risk for commercial aviation in the United States declined by 83%.⁽²⁾ This successful risk reduction resulted from systematic improvements to aircraft,

flight procedures, and air traffic procedures, mostly credited to rigorous data collection and analysis of both accidents and smaller incidents. **Accidents are obvious failures that for this industry are defined as events that involve fatalities and/or significant damage to the aircraft**. Consistent with organizational science literature, these accidents trigger investigations and motivate both learning and change.^(3–6) Adopting the International Civil Aviation Organization definition, **incidents are occurrences**, other than accidents, **associated** with the **operation** of an aircraft that affected or could have **affected the safety of operation**.⁽⁷⁾ Thus, **incidents are near-misses**, outcomes that **could have been worse** but for the intervention of **good fortune**.^(8,9) Our prior research has shown that some types of near-misses (i.e., some

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incidents) are harder to learn from than others.^(9,10) We explore this challenge for the airline industry.

March *et al.*⁽¹¹⁾ explain the difficulty in learning from near-misses. As they note, a **near-miss** between two planes in mid-flight offers observers **two distinct interpretations**—one that provides evidence for hazard **threat** and one that provides evidence for **system resiliency**.

Every time a pilot avoids a collision, the event provides evidence both for the threat [of a collision] and for its irrelevance. It is not clear whether the... organization **came [close]** to a disaster... or that the disaster was **avoided**. (p. 10)

The challenge of learning from near-miss incidents is that they do not **always evoke images of danger**, disaster, or feelings of systemic vulnerability. Bier and Mosleh⁽¹²⁾ demonstrate that **accident precursors** that are **recognized** as such will **provide** decisionmakers with **evidence of risk**, but there is always an **interpretation problem**. Near-misses may masquerade as success, and **apparent success tends to breed complacency** because decisionmakers institutionalize established organizational practices and routines and reduce organizational search activities aimed at identifying further system improvements.^(13–15) Consequently, prior perceived successes are often associated with reduced levels of future change in organizations,^(3,16) which leads toward **stagnation of organizational performance, and not toward learning and continuing improvement**.⁽⁶⁾ Yet, if some apparent successes are really near-misses (good outcomes because of good fortune in a complex stochastic process) then the institutionalized routines need to be questioned.

Recall that on March 5, 2000, Southwest Flight 1455 crashed through a 14-foot-high metal blast fence at the departure end of Runway 8 at Burbank, CA, continuing past the airport boundary, crossing a street, and coming to a stop near a gas station. There were 142 people on-board; two were seriously injured, 41 passengers plus the captain received minor injuries, and the airplane was substantially damaged.⁽¹⁷⁾ Following this accident, airlines learned. They traced the crash to the instability of the approach that required a descent that was too steep and required too much velocity. Most air carriers then developed procedures and **trained crews to avoid unstable approaches**. Moreover, if such approaches occurred, crews were to respond by executing a missed approach.⁽¹⁸⁾ However, at the time of the accident, the steep and fast approach at-

tempted by Flight 1455 was not unique among flight crews in general and not unique for flight operations at Burbank (though data are not available from that time to determine how common it may have been at Burbank). Investigators found that the pilots of Flight 1455 thought they had the situation under control up until touchdown because **“they were using the same techniques** to control their descent profile that they had used on previous ‘slam-dunk’ approaches which all worked successfully.”^(17, p. 68) The first officer told investigators that he had **previously seen captains successfully perform landings** “when the airplane had not been completely configured at 1,000 feet, when it had been ‘out of the slot’ at 500 feet, and when idle thrust was required on final approach.”^(17, p. 75) Burbank air traffic controllers interviewed by investigators after the accident reported that they had “seen airliners ‘make it’ down to a landing on runway 8 from as high as 7,000 feet or as fast as 260 knots ground speed over [the] Van Nuys [navigational aid].”^(17, p. 76) All of these prior near-misses, where risks were taken without negative consequence, deter any search for new routines. On the contrary, these **near-misses often reinforce dangerous behavior with the perception that this behavior effectively neutralizes hazards**.

This case is a clear example of **“normalization of deviance,”**⁽¹⁹⁾ in which a risky behavior becomes commonplace over time because there are no apparent negative consequences. In fact, Runway 8 at Burbank is relatively short for a Boeing 737 (6,032 feet of paved surface), and investigators believe that had the runway been longer, the “accident would probably not have happened even if everything else was the same.”^(17, p. 78) Had the runway been longer and no accident occurred, it is unlikely that the resulting near-miss **would have triggered the procedural change**. Yet, for airlines to continue to improve safety, the industry needs to attend to the yet undiscovered or **unrecognized risks** in the system without waiting for an accident to bring attention to them.

Fortunately, the industry acknowledges the importance of near-misses. Since the late 1990s, airlines and the Federal Aviation Administration (FAA) have placed a significant emphasis on **voluntary data-gathering programs** that enable airlines and government regulatory agencies to **spot** and correct **problems** before they lead to accidents. One such data-collection system, the Aviation Safety Reporting System (ASRS) administered by the National Aeronautics and Space Administration, receives more than 50,000 voluntary reports of safety

incidents and situations each year from pilots, air traffic controllers, cabin crew, dispatchers, and maintenance technicians,⁽²⁰⁾ and on March 21, 2012, the ASRS processed its one-millionth safety incident report.⁽²¹⁾ Since the mid 2000s, airlines and the FAA have also increasingly been able to examine flight-recorded data (i.e., air speed, altitude, control positions, etc.) through the Flight Operational Quality Assurance (FOQA) voluntary safety program. The FAA's Aviation Safety Information Analysis and Sharing (ASIAS) system has 42 member airlines sharing data integrating both ASRS (i.e., voluntary reports) and FOQA data (i.e., recorded data) with other data sources such as the National Transportation Safety Board (NTSB) investigations to further drive down risk. For example, ASIAS is helping the FAA and industry stakeholders with better characterization and understanding of events that can increase system risk, such as missed approaches, runway overruns, rejected takeoffs, and activation/malfunction of autobraking systems.⁽²²⁾ Finally, as more and more of the "next-generation" air traffic control system becomes functional, the FAA has newer and better technologies available to automatically capture additional surveillance data; for example, the FAA is using enhanced radar to identify whenever two aircraft violate flight separation rules.⁽²³⁾

Despite this commitment of resources and the impressive collection of system data, the human problem of cognitive error remains. Specifically, as noted above, events that do not result in accidents are subject to varying interpretations—Are they evidence of systemic vulnerability or resilience? For example, Waldron *et al.*⁽²⁴⁾ model collision risk on the airport surface using one month of surveillance data available from four Airport Surface Detection Equipment, Model X (ASDE-X) equipped airports: John F. Kennedy International Airport (JFK), Hartsfield-Jackson Atlanta International Airport (ATL), Minneapolis-St. Paul International Airport (MSP), and Memphis International Airport (MEM). The model uses detailed surveillance-based analysis of surface movement to measure the number and characteristics of discrete interactions during aircraft taxi movements that represent preconditions for possible aircraft ground collisions. During the month examined, there were no taxiway collisions between two aircraft at any of the four airports, but there were situations where aircraft were considered to be within 3 seconds of possible collision. This situation occurred eight times at JFK (out of 148,883 aircraft interactions), six times at MSP (out of 64,607 aircraft

interactions), five times at MEM (out of 16,433 interactions), and 21 times at ATL (out of 253,360). The resulting rates show that aircraft are in near collision conditions at ATL 31% more often than at MSP. Should this be interpreted as meaning that ATL operations are at greater risk for a collision in the future if current practices continue or that ATL operations are more skillful at having aircraft close together without problems?

These alternative interpretations make learning from near-miss incidents challenging. Learning requires that people notice a danger associated with nonaccidents that could produce accidents under different conditions. This can be challenging when decisionmakers are motivated to interpret near-misses as evidence that the system is working and when obvious signs of failure are absent.^(9,25,26) People are naturally prone to outcome bias—the general tendency to overweight observed outcomes and underweight the role of chance in producing said outcomes.⁽²⁷⁾ Thus, when decisionmakers interpret near-misses as successes, their confidence in the adequacy of organizational safety processes increases, such that they perceive fewer risks than they did previously.⁽⁹⁾

However, depending on the characteristics of the near-miss, sometimes these events can activate thoughts of system vulnerabilities. Tinsley *et al.*⁽¹⁰⁾ show that near-misses embedded with cues of obvious danger evoke feelings of caution and thoughts of system vulnerability, which counter the basic outcome bias effect and encourage greater scrutiny of near-misses. Their observation is consistent with Bier and Mosleh's⁽¹²⁾ analysis of precursor events where, for example, the U.S. Nuclear Regulatory Commission has chosen to limit the use of the term "precursors" to events that exceed a specified level of severity, and that those serious events do encourage greater scrutiny.

The challenges in building a useful incident database are: (1) understanding when near-misses will be recognized as something different from successes and logged into the system, and (2) understanding when near-misses will be attended to because they activate thoughts and feelings of danger, which can motivate learning. Here, we investigate the second challenge for the airline industry. Using an empirical analysis of archival data from the U.S. commercial aviation industry from 1990 to 2007, we test the main effects and interaction of organizational experience with both near-misses and accidents. We note the first challenge, though, to remind

readers that the incident database we examine is a partial estimate of the true frequency of near-misses because it considers only those events that were included in the database (i.e., identified as incidents or accidents).

2. THE MODEL

We examined all U.S. commercial airlines that operated from 1990 to 2007 and were considered by the U.S. Department of Transportation (DOT) to be “large certificated air carriers” (which includes all airlines with annual operating revenues of \$20 million or more). Each airline in the sample was observed quarterly (with some entries and exits over time), creating an unbalanced panel data set. The original sample included 118 airlines. However, due to lags required to construct some variables (described below), airlines that operated for less than three years during the sampling period were excluded from the sample. A total of 54 airlines were excluded based on this requirement, leaving 64 airlines and 2,955 quarterly observations of these airlines in the sample.

We utilized three data sources to construct the sample. Data on quarterly airline operations were drawn from the DOT’s Bureau of Transportation Statistics Air Carrier Statistics and Air Carrier Financial Reports database. We collected data on airline accidents from the U.S. NTSB aviation accident database. Data on airline incidents (near-misses) came from the FAA Accident/Incident Data System. We used the Causal Model for Air Transport Safety (CATS) developed for the Netherlands Ministry of Transport, Public Works and Water Management⁽²⁸⁾ to categorize accidents and incidents. The CATS model was developed to systematically examine the challenges in the air transport industry with a goal to “identify areas for improvement to the technical and managerial safeguards against accidents.”^(28, p. 7)

2.1. Dependent Variable

The purpose of this study is to examine organizational learning from experience with near-misses and accidents. Although, strictly speaking, organizational learning can be defined as a change in organizational routines⁽²⁵⁾ or individuals’ cognitive structures,⁽²⁹⁾ it is standard practice in the literature on organizational learning from accidents to operationally define learning as a change (typically an improvement) in performance.^(29,30) We adopt this convention here

by defining learning as a modification in an organization’s accident rate. In other words, an organization will be said to have learned to the extent that its future accident rate declines. Although organizational learning is not the only reason that accident rates may decline over time, we expect this operational definition to be a reasonable assumption here as we will attempt to statistically control for alternative explanations (discussed below). This assumption is also made in prior work on airline learning.^(30,31)

Consequently, the dependent variable used in all analyses was a quarterly accident count—the number of accidents experienced by an airline in a given quarter. We adopt the NTSB’s definition of an aviation accident as an event that takes place during the operation of an aircraft that causes the death or serious injury of a person or causes structural failure or significant structural damage to the aircraft.⁽³²⁾ We labeled the dependent variable *Accident Count*.

2.2. Independent Variables

The independent variables used in the analysis are event counts associated with prior accidents and prior near-misses. First, we considered *Prior Accidents* as a count of the number of accidents experienced by a given airline over the three years (12 quarters) prior to the current quarter. We adopted this three-year window for learning from accidents following prior work on organizational learning from aviation accidents that found that airlines learn primarily from accidents occurring within the past three years.⁽³³⁾ Consequently, in all reported analyses, the variable *Prior Accidents* represents a count of the number of accidents experienced by a given airline over the prior three years. We will use this *Prior Accidents* variable to test our first hypothesis that airlines learn from accidents, both their own accidents and others. Note, the results are not dependent on examining accidents occurring within the prior three years, and qualitatively identical results are obtained when windows of between two years and four years are used to construct the prior accident counts.

We constructed additional independent variables dealing with airline near-misses. We drew near-miss data from the FAA’s Accident/Incident Data System, which contains information on investigated incidents experienced by U.S.-based airlines. The database includes events that take place during the operation of an aircraft that fail to qualify as an accident, but that could affect the safety of the aircraft,

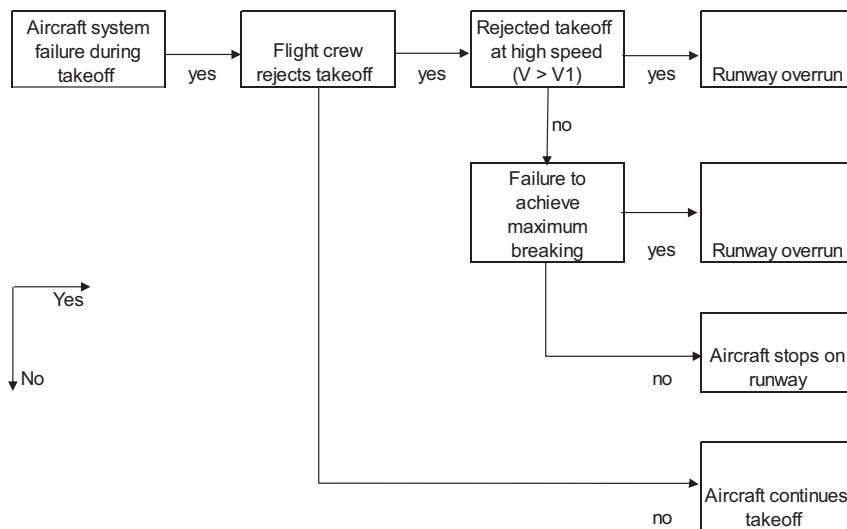


Fig. 1. Example of ESD from CATS model.^(28, p. 27)

including events that caused minor damage to an aircraft and events that caused one or more minor injuries, as well as events that resulted in no damage and no injuries but during which airline personnel identified a hazardous condition. Note, engine failure or damage to landing gears, wheels, tires, flaps, brakes, and wing tips are not considered substantial damage.

We define four variables to serve as proxies for examining which near-miss events are recognized as dangerous system events. We used the CATS framework⁽²⁸⁾ that describes ways that aviation processes can go awry. This framework defines incidents and maps these events into event sequence diagrams (ESDs) to categorize how these events can lead to any anomaly (both incidents and accidents). For example, one ESD considers an aircraft system failure during takeoff and depending on pivotal events (flight crew rejecting or not rejecting the takeoff, maximum braking achieved or not achieved, etc.), different results are considered: a runway overrun or a successful stop on the runway. This ESD is shown in Fig. 1. In the ESD, a runway overrun could occur with significant injury and damage (as in the Southwest Flight) but a runway overrun could occur when an aircraft rolls into grass at the end of the runway with no damage. Thus, both types of events (accidents and near-misses) could be associated with the same ESD category because the category depends on the initiating event, has multiple possible outcomes based on pivotal events, and does not specify a level of outcome damage.

Table I. CATS Event Sequence Diagrams

CATS Events Sequence Diagram	Number of Events	≥1 Accident
None	320	n.a.
Aircraft failure during takeoff	18	
ATC event during takeoff	8	X
Directional control inappropriate takeoff	6	X
Directional control failure takeoff	11	
Incorrect configuration during takeoff	1	X
Takeoff with contaminated flight surface	1	X
Single engine failure during takeoff	40	X
Fire onboard aircraft	48	X
Flight control system failure	75	X
Flight crew incapacitation	2	
Ice accretion on aircraft in flight	5	X
Airspeed, altitude, attitude display failure	12	
Adverse weather	201	X
Single engine failure in flight	51	
Unstable approach	114	X
Wind shear during approach or landing	1	
Handling inappropriate during flare	3	X
Handling inappropriate during landing roll	7	
Directional control failure landing	59	X
Aircraft on collision course in flight	34	
Runway incursion involving a conflict	11	X
Cracks in aircraft pressure boundary	18	
Conflicts with terrain	3	X
Conflicts on taxiway or apron	175	
Wake vortex encounter	1	
Loss of control due to poor airmanship	3	

Table I describes the relevant CATS ESDs with associated counts of events in our data set and notes whether or not that event category was ever associated with a major accident in our data set.

Note, 26% of the events in our data set did not have a corresponding CATS ESD. Missing scenarios include, for example, fumes/odor (not resulting from fire), nuisance warnings, and false alarms. But it is important to remember that even fumes, nuisance warnings, and false alarms in commercial aviation can cause abrupt maneuvers and unnecessary evacuation, which can each lead to injuries or aircraft damage. Other missing scenarios included problems on the ground before takeoff or after completion of the landing roll, such as landing gear failures during taxiing and problems with the auxiliary power units (APUs) at the gate, which can be misinterpreted as fire-related events. Additionally, no personal injury scenarios on the ground are included, such as the flight attendant getting a hand caught in the aircraft door or a passenger or ground crew making fatal contact with a turbo propeller blade. So even when resources and effort are expended to analyze and study accident scenarios as was done developing the CATS model, in a complex system such as commercial aviation, it is difficult for all possible scenarios to be addressed.

Our first pair of variables, *Near-Miss Identified Scenario* and *Not Near-Miss Identified Scenario*, are defined by whether or not a CATS-relevant ESD was developed or not for that event in our data set. *Near-Miss Identified Scenario* is a count of the number of near-misses an airline experienced over the prior three years for which a CATS-relevant ESD exists, and *Not Near-Miss Identified Scenario* is a count of all other near-misses experienced by the airline over the same time period for which a CATS-relevant ESD did not exist. These two variables will be used to test our second hypothesis that airlines learn from events that they recognize and study as potential failure modes of operation and that they do not learn from those that are not a focus. And again, we use the CATS model⁽²⁸⁾ as a proxy for identifying this set of focal events.

For our second pair of proxy variables, we further divided the *Near-Miss Identified Scenario* (i.e., CATS recognized) category into: events that had at any time in our data set resulted in a major accident and those that had not, where a major accident meant that either the aircraft was destroyed and/or the event resulted in fatalities. Therefore, our variable, *Similar Category Major Accident*, is a count of near-misses experienced by an airline over the prior three years that were in categories of events that had resulted in at least one major accident in our data set. Our final near-miss variable, *Not Similar*

Category Major Accident, is a count of other near-misses for which a CATS-relevant ESD existed, but were in categories that had not resulted in major accidents (again the count is over the prior three years). These final two variables (in conjunction with the *Not Near-Miss Identified Scenario* variable) will be used to test our third hypothesis that airlines only learn from those events recognized as potential failure modes of operation by the CATS framework and within those categories of events only those that are recognized as serious, dangerous events (i.e., where the possible bad outcome is salient). As with prior accident counts, the reported results with regard to both versions of the near-miss variables are not contingent on using a three-year window for counting prior near-misses; the results were qualitatively unchanged when the near-miss counts were constructed using windows of any increment between two years and four years.

2.3. Control Variables

We also included several control variables in the analysis to account for factors other than accident and near-miss experience that might impact future airline accident rates. First, to control for exogenous changes that may have occurred over time, we included year fixed effects in the analysis. Second, we included the number of accidents experienced by other airlines over the previous three years (12 quarters) to account for vicarious organizational learning. This count of others' prior accidents included accidents experienced by all U.S.-based "large certificated air carriers" other than those experienced by the focal airline, and thus represent virtually the entire U.S. commercial aviation industry. Third, because large airlines are at a higher risk for accidents (because of more flights per day) than small airlines, we also controlled for airline size in all models by including the number of airline departures in a given quarter. Because most airline accidents occur during takeoff or landing,⁽³⁴⁾ controlling for airline departures effectively removes the influence of airline size on accident rate.

Although most accidents occur during takeoff or landing, some accidents also take place during flight. Thus, to control for this marginal additional risk on longer flights, we also controlled for airlines' average stage length (the average number of miles flown by an airline per departure). Moreover, to control for the effect of airline capitalization on accident rate, we included the ratio of airline assets to departures

as a control variable. We also included airline operating margin, calculated as (operating income/net sales) $\times$ 100, to control for the effect of profitability. Finally, we included an indicator variable in the analysis to account for whether or not an airline was undergoing bankruptcy reorganization (a relatively common occurrence during our sampling frame) in a given quarter to control for the possibility that such airlines are at elevated accident risk (1 = undergoing bankruptcy reorganization, 0 = not undergoing bankruptcy reorganization). Airline accident rates may decline for reasons other than learning by individual airlines, especially if government regulations change or technological advancements increase safety across the industry. To control for these factors, we also include fixed year effects in all models. Although these year effects are unlikely to account for all such effects that vary over time (for example, different airlines may adopt new technologies at different times), they should account for the vast majority of regulatory and technological changes as these changes impact all airlines in the industry at the same time.

Table II reports summary statistics and pair-wise correlations for all study variables. The table reports the average number of accidents and near-misses of various types across the sample period. The number of accidents and near-misses experienced by airlines in the sample also vary somewhat over time, generally trending downward, but with quite bit of noise. The highest accident rate in the sample is 1.82 accidents per million departures in 1998, and the lowest is 0.80 accidents per million departures in 2004. The number of near-misses peaks at 1.74 near-misses per million departures in 1993 and troughs at 0.71 near-misses per million departures in 2003.

To test whether multicollinearity among the explanatory variables could adversely affect the analysis, we calculated the variance inflation factor (VIF) for each independent and control variable as well as the eigenvalues for linear combinations of independent and control variables based on the correlation matrix. The largest VIF for any of the explanatory variables was 2.99 (for total departures), which is well below the traditional cutoff of $VIF > 10$ to indicate harmful collinearity.⁽³³⁾ Similarly, the largest condition index for any linear combination of explanatory variables based on their eigenvalues was 8.8, again well below the traditional cutoff of condition index >30 to suggest collinearity. Thus, we concluded that the results of our analysis would not be adversely impacted by multicollinearity.

3. ANALYSIS AND RESULTS

The dependent variable of interest here is an event count—airline accident count. The Poisson distribution describes random events that occur independently over time and is the standard approach to modeling airline accidents.^(35–39) Use of the Poisson model requires the assumption that the mean and variance of the count variable are equal (the equidispersion assumption). When a count variable's variance is greater than its mean (overdispersion), this assumption is not met and the Poisson model is inappropriate. A likelihood ratio test comparing the Poisson models used in this study to more general negative binomial models failed to reject the equidispersion assumption. Therefore, Poisson regression was used in all analyses. To confirm that reported results are not driven by this specification decision, we also ran all reported models using negative binomial regression, which does not require any assumptions about the relationship between the sample's mean and variance. The results of this supplemental analysis were essentially identical to those reported below. This finding lends confidence to the reported results.

Because the sample contained repeated observations of the same airlines over time, the data were clustered by airline. To control for the possibility that such clustering might bias the regression results, we included fixed airline effects in all models. The Poisson model we estimate is as follows:

$$P(y_{it}) = \frac{\exp(-\lambda_{it}) \times \lambda_{it}^{y_{it}}}{y_{it}!}, \quad (1)$$

$$\begin{aligned} \ln(\lambda_{it}) = & \beta_t + \beta_{acc} * \text{Prior Accidents} \\ & + \beta_{nm} * \text{Prior Near - Misses} \\ & + \beta_{con} * \text{Controls} + \mu_i + \varepsilon_{it}, \end{aligned} \quad (2)$$

where i indexes airlines, t indexes years, and λ is the Poisson parameter. β_t is a vector of intercepts for each year, β_{acc} is a coefficient for prior accidents, β_{nm} is a vector of coefficients corresponding to the various types of near-misses discussed above, β_{con} is a vector of coefficients corresponding to the control variables, μ_i is a vector of airline fixed effects, and ε_{it} is the error term. We operationalized this model using the `xtPoisson` command in Stata 13 with airline fixed effects specified and a series of dummy variables accounting for year fixed effect. The use of airline fixed effects in the analysis, although important to control for unobserved differences among airlines, required the omission of airlines from the analysis

Table II. Summary Statistics and Pair-Wise Correlations

Variable	Mean	SD	1	2	3	4	5	6	7	8	9	10	11
Dependent variables													
1. Accident count	0.13	0.68											
Independent variables													
2. Prior accidents	1.46	3.02	0.49										
3. Near-miss identified scenario	0.95	1.94	0.41	0.71									
4. Not near-miss identified scenario	0.54	1.22	0.38	0.64	0.67								
5. Similar category major accident	0.64	1.43	0.37 ^a	0.650.91	0.59								
6. Not similar category major accident	0.31	0.86	0.31	0.51	0.73	0.53	0.39						
Control variables													
7. Others' prior accidents	90.55	22.00	-0.04	-0.04	-0.13	-0.15	-0.11	-0.10					
8. Departures (1,000,000s)	0.11	0.17	0.48	0.78	0.72	0.67	0.70	0.46	-0.11				
9. Ave. stage length (miles)	958.40	791.32	-0.04	-0.06	-0.07	-0.04	-0.04	-0.08	0.10	-0.17			
10. Assets per departure (\$100,000s)	1.00	17.29	-0.01	-0.02	-0.02	0.00	-0.02	-0.02	0.00	-0.03	0.11		
11. Operating margin, $t-1$	-0.10	3.38	0.01	0.01	0.02	0.01	0.02	0.01	-0.02	0.05	0.00	0.01	
12. Chapter 11 bankruptcy protection	0.04	0.19	0.02	0.06	0.03	0.03	0.00	0.07	0.05	0.05	0.01	-0.01	-0.01

Note: $n = 2,955$ airline-quarters. Pair-Wise correlations were calculated across all observations in the sample (observations of 64 airlines across a varying number of quarters), and thus represent covariance across airlines and over time.

that did not experience an accident during the sampling period because those airlines had no variance on the dependent variable. This restriction reduced the number of airlines in the analysis to 50 and the number of observations to 2,158.

Table III reports maximum-likelihood estimates for the fixed effects Poisson regression analysis of quarterly airline accident counts. Model 1 includes only the control variables, of which only airline departures are significantly related to airline accidents. Model 2 introduces airline prior accident count, which is negatively and significantly related to accident rates, indicating that airlines learn from their own accidents to reduce the likelihood of experiencing future accidents, thus supporting our first hypothesis. This effect is not only statistically significant, but also practically significant. The incident rate ratio (IRR) for prior accidents in Model 2 is 0.93, which indicates that one additional accident that an airline has experienced in the prior three years reduces the likelihood of that airline experiencing an accident in the current year by 7%.

Model 3 introduces the counts of near-misses that fit into CATS-relevant ESDs and those that do not. Neither of these variables reaches statistical significance, indicating that airlines do not learn more from near-misses that fall into ESDs identified in the CATS model than from those that do not, thus *not* supporting our second hypothesis. These data show no evidence that airlines are learning from *all* past incidents recognized as potential failure modes.

Model 4 introduces counts of near-misses distinguished by whether their category identified by the CATS model had previously resulted in a major accident or not, and also retains the count of near-misses that fall outside of the CATS model. The *Similar Category Major Accident* near-miss count was negatively and significantly related to airline accident rate, while the *Not Similar Category Major Accident* near-miss and *Not Near-Miss Identified* counts were not related to accident rate. These results indicate that airlines learn from near-misses that fall into categories of events where danger is salient, but not from the other category of near-misses, thus supporting our third hypothesis. The effect of the *Similar Category Major Accident* near-miss count was practically significant in addition to being statistically significant. Its IRR was 0.92, indicating that an airline that experienced one additional near-miss in a category where danger is salient during the prior three years was 8% less likely to have an accident in the current year than it would have been without that additional near-miss.

Finally, Model 5 is the full model and includes all of the variables simultaneously (with the exception of *Near-Miss Identified Scenario*, which is an alternate variable based on the same near-miss events as the *Similar Category Major Accident* and *Not Similar Category Major Accident* variables and, consequently, cannot be included in the same model). With respect to the independent variables, the results in this model are similar to those discussed above, further supporting our first and third hypothesis.

Our sample included some airline-quarters with abnormally high numbers of accidents and near-misses (such as the third quarter of 2001, which contained the September 11 attacks). To verify that the reported results were not driven by outliers, we winsorized the top 5% of observations in the sample and reran the analysis.⁽⁴⁰⁾ With regard to all independent variables and hypothesis testing, the results of this robustness check were the same as those in the reported models. Thus, we concluded that outliers were not unduly influencing the results.

4. DISCUSSION

The results presented here are largely in line with prior work on organizational learning and near-misses. Specifically, airlines learned to improve their safety performance in response to their own accidents and accidents experienced by other airlines. Also, we see that airlines can learn from some types of near-misses—those that were of the same category of event that had at other times resulted in major accidents and thus could be associated with clear signs of danger—but not from near-misses that lack this association. Thus, airlines learn from near-misses, but only those associated with obvious signs of risk.

This pattern of findings suggests that U.S. commercial airlines have the capacity to improve their safety performance based on near-misses, but that this learning does not occur automatically following all near-misses. Thus, the capacity of airlines to use near-miss experiences to improve safety may not be fully realized. U.S. airlines may be able to further enhance safety if they could learn to view even not obviously threatening near-misses as learning opportunities and broaden the reporting and use of such events by flight crews, ground crews, air traffic controllers, maintenance personnel, and other stakeholders. Increased recorded and surveillance data will also help if nonthreatening near-misses can be correctly identified and interpreted in these data.

Table III. Fixed Effects Poisson Regression Models of Airline Accidents

	Model 1	Model 2	Model 3	Model 4	Model 5
Independent variables					
Prior accidents		-0.07* (0.02)			-0.07** (0.02)
Near-miss identified scenario			-0.04 (0.03)		
Not near-miss identified scenario			-0.01 (0.03)	-0.03 (0.04)	-0.02 (0.04)
Similar category major accident				-0.09* (0.03)	-0.08* (0.03)
Not similar category major accident				0.04 (0.04)	0.03 (0.04)
Control variables					
Fixed year effects	Included	Included	Included	Included	Included
Fixed airline effects	Included	Included	Included	Included	Included
Others' prior accidents	0.00 (0.01)	-0.03† (0.02)	0.00 (0.01)	0.00 (0.01)	-0.03 † (0.02)
Departures (1,000,000s)	1.41† (0.85)	1.66† (0.90)	1.46† (0.86)	1.43† (0.89)	2.01* (0.93)
Ave. stage length (miles)	-0.01 (0.01)	-0.01 (0.01)	-0.01 (0.01)	-0.01 (0.01)	-0.01 (0.01)
Assets per departure (\$100,000)	-0.19 (0.65)	-0.20 (0.69)	-0.22 (0.73)	-0.19 (0.65)	-0.21 (0.71)
Operating margin, t - 1	0.89 (0.56)	0.85 (0.56)	0.86 (0.56)	0.92 (0.54)	0.88 (0.54)
Chapter 11 bankruptcy protection	-0.25 (0.25)	-0.32 (0.26)	-0.34 (0.26)	-0.31 (0.26)	-0.39 (0.26)
N	2,158	2,158	2,158	2,158	2,158
Log likelihood	-883.68	-879.11	-882.40	-879.76	-875.64

Note: Standard errors are in parentheses.

† $p < 0.1$; * $p < 0.05$; ** $p < 0.01$. Two-tailed tests.

To do this, we suggest two steps, based on our findings here as well as prior work on near-misses and organizational learning:

(1) *Continue the successful data-collection efforts, but continue to broaden what is reported.* As noted in our study, no learning can occur if the data are not collected. We could only examine a partial set of near-miss events, that is, those that were reported. Near-miss reporting systems (such as ASRS) used in commercial aviation are the envy of other industries such as medicine and cyber security where near-miss reporting is much less developed, and these efforts and the vigilance associated with them must be continued. The challenge is to expand the identification and reporting of all types of near-misses, but at the same time keep the cost of reporting to a minimum.

Without priming, people tend not to think through the possible negative consequences of near-misses,⁽⁹⁾ and thus do not recognize many of these events as candidates for reporting to systems like ASRS. Relatively simple changes in the training

and communications that airline personnel receive may be enough to prompt flight crews and other stakeholders to recognize the possibility of alternative outcomes rather than the successful outcome that occurred. Additionally, efforts must continue to keep the reporting easy and the costs low. As Paté-Cornell⁽⁴¹⁾ describes: "A balance has to be found between the time necessary for an appropriate response, the corresponding benefits, and the cost of false alerts, in terms of both money and human reaction."

Near-miss reporting systems in the aviation industry will continue to benefit from the amount of automated data available from the "next-generation" air traffic control investments.⁽²²⁾ As more recorded and surveillance flight data become available, these additional data will complement the voluntary reporting data if used to further identify near-misses where a dangerous outcome was not salient, and may also come at lower cost than data that rely on aviation personnel to report.

(2) *Remain vigilant toward any deviations from normal and uncover root causes of the deviations.* Our findings indicate that airlines learn from near-misses that fall into categories of events where danger is salient, but not from the other category of near-misses. Too often when events are identified as near-misses, if the effects of the near-miss can be easily corrected, decisionmakers may assume that by correcting the outward signs of the near-miss they have eliminated the problem, and thus danger does not seem salient. Van der Schaaf and Kanse,⁽⁴²⁾ in a study of workers in the chemical industry using daily diaries, found that one of the most common reasons workers did not report a self-error was that the mistake had no consequences. Decisionmakers in an organization must consider an event to be some sort of anomaly before it will be catalogued. When latent errors that generated the near-miss are not identified and corrected with revised procedures and improved checklists, future near-misses will almost certainly recur, and catastrophe can follow if those latent errors interact in novel ways with other hazards and errors. Airline personnel should be trained to recognize “normalization of deviance”⁽¹⁹⁾ (i.e., events that were initially considered to be unacceptable but that occur so often that—over time—they come to be expected and then accepted as normal, such as the rapid descent landings into Burbank airport) and to be concerned if they detect this attitude.

Ensuring that government and airline safety personnel search for the root causes of all deviations, understand the root causes, and make appropriate changes to procedures and checklists will be necessary to continue to see further improvements in safety in the U.S. system in current conditions when serious accidents are rare. Steps to support identifying root causes are occurring in the aviation industry, including the development of detailed fault tree models for each node in the CATS ESD models⁽²⁸⁾ and a U.S. version of the CATS model integrated into the FAA’s Accident Investigation and Prevention Division’s Integrated Safety Assessment Model.⁽⁴³⁾ These efforts to understand the root causes of accidents need to continue.⁽⁴⁴⁾

5. CONCLUSIONS

After decades of significant safety improvements, U.S. commercial airlines still face ongoing challenges from an operating environment of financial instability in the airline industry, security threats, evolving cockpit automation, and new

air traffic control procedures and systems being introduced to accommodate growing density of air traffic. Also, research on aviation safety finds that pilots or other crew members make at least one potentially hazardous error on 68% of commercial airline flights, but very few of these errors lead to an accident⁽⁴⁵⁾ because few of these errors interact with other errors and hazards in ways that existing safety systems have not anticipated.^(46,47) When such errors do occur, but do not generate an accident, they should be considered near-misses. The difference between a near-misses and a larger failure may only be good fortune, and thus learning from near-misses, in addition to learning from failures that actually materialize, should reduce the likelihood that the aviation industry would experience major failures in the future. Thus, recognition of near-miss events has the potential to allow decisionmakers to learn how to reduce the likelihood of significant accidents in the future, above the learning that occurs from accidents themselves.

Obviously, in practice, not every deviation will be noticed or recorded, making every near-miss database a partial estimate of actual abnormalities. Whether conscious or not, people will trade off the costs of cataloguing incidents relative to their potential learning benefit. Our research suggests that the potential learning benefit is chronically underestimated, as often it is not until several incidents are reported that a pattern may emerge or that luck may run out and a major accident occurs. To combat the underestimation of value of learning from near-miss events, we strongly advocate that costs of near-miss detection and reporting be minimized. Rather than expending resources to try to determine, *a priori*, where the cost/benefit line is (i.e., which near-misses simply are too trivial to report), we suggest spending these resources to lower data-collection costs. What may seem to be “over-collection of near-misses” today could turn out to be valuable data in the future.

The work reported here demonstrates that U.S. airlines have been adept at learning to improve safety from accidents and from near-misses that are associated with a clear cue of danger. Our lingering concern is the many near-misses that are not associated with prior accidents do not appear threatening enough for airlines to attempt to learn from them. In order to continue to further improve safety, airlines need to institute policies that encourage learning from seemingly innocuous near-misses. The U.S. aviation system has achieved truly astounding levels of safety. But such success has, ironically, all but eliminated

the most obvious sources of learning for additional improvement. Future increases in system safety will require learning from ever smaller and less obvious near-misses.

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